

In-work policies in Europe: Killing two birds with one stone?

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Abstract

Generous social assistance has been held responsible for inactivity traps and social exclusion in several European countries, hence the recent trend of promoting employment through in-work transfers. Yet, the relative consensus on the need for ‘making work pay’ policies is muddled by a number of concerns relative to the design of the reforms and the treatment of the family dimension. Relying on EUROMOD, a EU-15 integrated tax-benefit microsimulation software, we simulate two types of in-work benefits. The first one is means-tested on family income, in the fashion of the British *Working Family Tax Credit*, while the second is a purely individualized policy. Both reforms are built on the same cost basis (after behavioral responses) and simulated in three European countries suspected to experience large poverty traps, namely Finland, France and Germany. The potential labor supply responses to the reforms and the subsequent redistributive impacts are assessed for each country using a structural discrete-choice model. We compare how both reforms achieve poverty reduction and social inclusion (measured as the number of transitions into activity). All three countries present different initial conditions, including existing tax-benefit systems and distribution of incomes and wages. These sources of heterogeneity are exploited together with different labor supply elasticities to explain the cross-country differences in the impact of the reforms.

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1. Introduction

Poverty has been reduced in many industrialized countries by the development of large-scale welfare systems which include generous social assistance schemes for the poorest. However, there is a well-known risk that the instruments used for this purpose generate social exclusion by making work financially unattractive, especially to less productive workers. Consequently, benefit schemes must be designed in a way which preserves sufficient income while ‘making work pay’. In this respect, pioneering in-work transfers have been introduced for many years in the US and the UK, with the Earned Income Tax Credit (EITC) and the Working Family Tax Credit (WFTC) respectively. Several European countries have recently experienced the implementation of similar reforms or actively debated about doing so.

Yet, the relative consensus on the need for this type of policy is muddled by concerns about efficient design, given the framework conditions and the general objectives pursued. In particular, the treatment of the family dimension is a crucial issue which deserves specific attention. EITC/WFTC-type of policies, means-tested on household income, are suspected to increase work incentives for single workers but to discourage married women with working partner. It is typically argued that this efficiency loss is outweighed by the equity gains from targeting families in need. In contrast, individualized schemes appear to combine clearer incentive effects with less efficient targeting, since resources may be transferred to workers in well-off families. Overall, whether redistributive and efficiency objectives can be reconciled in a single policy measure remains an open question.

This paper provides the first comprehensive comparison of these two alternative policies and their implications on labor force participation. To do so, we simulate the labor supply effects of a purely individualized in-work transfer and of an extended version of the WFTC. We conduct these simulations for three European countries which experience inactivity traps, namely Finland, France and Germany. Despite numerous debates (esp. in France) on the appropriateness of in-work transfers, none of these countries has implemented policies as generous as the EITC/WFTC. More generally, we tackle the issue of the design of in-work policies by exploring for each country the role of initial conditions, e.g. the institutional framework, the distribution of income, etc. These conditions are key to the success of a reform, as stressed in previous studies (e.g. Bertola, 2000), but have not received systematic attention in large-scale analyses.

To analyse the potential effects of each reform on incentives and redistribution, we estimate a structural labor supply model for single and married women, using micro data for the three countries and a tax-benefit microsimulation program. To perform meaningful comparisons, the reform scenarios are calibrated to reach the same budgetary cost after potential behavioral responses.

To clarify policy analysis, we compare the reforms in the light of two clear-cut policy goals, namely *poverty reduction* and *social inclusion* (measured as the number of transitions into work). Specifically, we question which of the suggested in-work benefits succeeds best on each account and whether the incentive effect of in-work transfers is significant in poverty reduction. Firstly, results show large efficiency gains from moving from family to individually based policies. While the impact on poverty is quite similar between the two policies, the individual wage subsidy increases women’s employment. At the same time, the negative employment effect of the WFTC on married women offset by far the positive effect on single women. This result comes in contrast to those in the US and the UK. Secondly, it turns out that the WFTC can be particularly effective when priority is assigned to lone mothers, a group showing the highest risk of poverty and social exclusion. We also discuss what can be achieved given each country’s social policy agenda.

Table 1
Social assistance per country

	France	Germany	Finland
No. of hh on SA	1,729,000	2,700,000	264,000
Proportion of hh on SA	7.52%	7.08%	11.21%
Social assistance (billions EUR)*	5.50	9.40	0.46
Social assistance (% GDP)	0.38%	0.44%	0.36%
Average yearly amount per hh (EUR)	3,181	3,481	1,755
% of hh with EMTR >70%	4.40%	5.90%	5.70%

National statistics for 2001: Federal Minister of Labor and Social Affairs (Germany), Social Welfare Board (Finland), Ministry of Social Affaires (France).

* This includes only direct cash transfers; in particular, it includes cost-of-living assistance but excludes assistance in special situations in Germany; it includes minimum pension.

The layout of the paper is as follows. In Section 2, we present the recent trend in in-work policies in Europe. Section 3 outlines the structure of two types of policies and details the assumptions regarding their design and their simulation. In Section 4, initial conditions are closely analyzed and provide useful insights to explain the results. For each country, predicted elasticities are compared with related findings in the recent literature. Section 5 analyses the potential effects of the reform on incentives and redistribution and suggests interpretations of the cross-country and cross-reform differences. Section 6 concludes.

2. In-work policies in Europe

2.1. Social assistance and in-work transfers

In-work policies have been in place for a long time in the US, with the EITC, and in the UK with the *Family Credit* and its successors. Canada, Ireland and New Zealand have also had a relatively long experience with such schemes. The official objective set forth by policy makers is twofold: (i) to expand employment by increasing work incentives, (ii) to increase income of disadvantaged groups (see [Pearson, 2002](#)). There are also country-specific objectives such as the reduction of child poverty in the UK. Such instruments are often seen as more politically acceptable than a rise in social assistance and more desirable given the fear of increasing work disincentives; they are also seen as more efficient than an increase in the minimum wage, which might push up wage rates above the market equilibrium.¹

In recent years, the idea of in-work transfers has been spreading in continental Europe too, especially in those countries characterized by large levels of social transfers which are suspected to discourage employment. There, ‘make work pay’ (MWP) policies have been suggested to recreate an attractive financial gap between welfare and paid work. In this respect, we focus in this study on Finland, France and Germany, three countries in which suspected disincentive effects are particularly marked due to generous welfare systems. [Table 1](#) presents the relative generosity of the three systems, varying around 0.4% of GDP.

In these three countries, income assistance transfers are computed as a basic minimum income reduced by total household resources and in particular by labor income. This differential structure corresponds to an implicit 100% taxation on earnings. In [Fig. 1](#), depicting the budget

¹ The role of wage floors is not explicitly modeled in the present exercise. See [Laroque and Salanié \(2002\)](#) for a more comprehensive set-up.

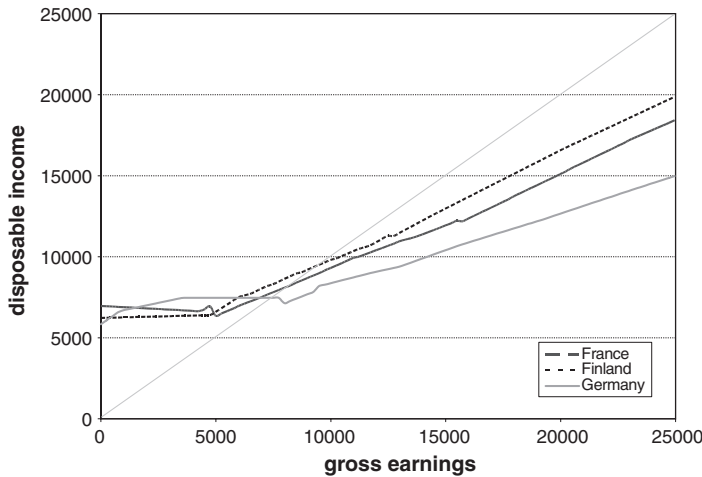


Fig. 1. Budget curves for a single individual in France, Germany and Finland.

curve of a single individual, this materializes in the flat portion of the curve for low levels of earnings. Interestingly, this feature is identical in all three countries. It is to be found for all family configurations with the exception of two-earner couples. Some national specificities are worth mentioning. In Germany, a lower implicit taxation applies but only for very low earnings. The level of social assistance is more generous there (4650 euros per year for a single individual, versus 4000 in France and Finland), which explains why the flat portion is larger than in the two other countries. Overall, the level of ‘safety net’ is very similar in all three countries, due to the fact that housing benefits play a larger role in France (3000 euros for an inactive person, i.e. 44% of her disposable income) and Finland (2000 euros, i.e. a third of her disposable income) than in Germany (1200 euros, 20% of her disposable income).²

Disincentive effects are often characterized in terms of effective marginal tax rates (EMTRs hereafter), measuring the implicit taxation of a marginal increase in earnings. Using micro simulation, we have computed EMTRs for active or potentially active households (i.e. households with the head aged between 25 and 60, neither disabled nor in full-time education nor retired) in each of the three countries. As shown in Table 1, a substantial proportion of households (between 4 and 6%) face EMTRs above 70%.³ This presumed ‘inactivity trap’ is common to the three countries while they present wide heterogeneity on other accounts, fully exploited in the following analysis.

2.2. Individual versus family-based schemes

Overall, a consensus seems to emerge on the need for MWP policies in Europe and on essential aspects of their design (see Duncan et al., 2003). This view is nevertheless muddled by important concerns regarding (i) the ability of such policy measures to achieve multiple targets

² For higher levels of income, divergences across countries in the curves are due to differences in income taxation and social security contributions (SSC). In addition, housing benefits in Finland and Germany are sharply phased out as earnings increase (they exhaust at a level of 12,500 euros of yearly gross income for a single individual, versus 15,500 euros in France).

³ This characterization of the inactivity trap is fully described in a companion paper (Bargain and Orsini, 2005) which closely compares the social assistance systems in the three countries and the distribution of EMTRs.

and (ii) the fact that there is no unique definition of a MWP policy. In particular, the treatment of the family dimension and the choice of the unit of assessment may strongly condition the results. Two broad groups of possible schemes are usually encountered, although hybrid measures also exist. On the one hand, some countries have introduced family-based measures, i.e. in-work transfers which depend on household size and are means-tested on family income. These reforms, whose best examples are the EITC and the WFTC, are known to be well-targeted at poor working families,⁴ to unequivocally encourages the participation of single individuals but also to discourage secondary earners in couples, bringing about a gender bias against the participation of women (see Eissa and Hoynes, 2004; Blundell et al., 2000, among others). The generosity of the reform also implies a high taper rate in the phase-out range, hence large increases in EMTRs and potential disincentives at the intensive margin.

On the other hand some countries like Belgium have shortly experienced purely individualized measures, conditioned on individual earnings only. Given a similar budgetary cost, this type of measure would imply smaller benefits and a larger number of recipients. Indeed, low-paid individuals in well-off families may well receive some transfers. This policy is thought of as having greater incentive effects than the family-based alternative as it has no discouraging effects on second earners in couples; less generous amounts may also imply smaller increases in EMTRs in the phase-out region. It seems that policy makers may have to choose one instrument or the other, depending on how they balance redistributive and incentive objectives. Things may be more complicated. For instance, job-enhancing policies can well be viewed as a way to reduce poverty through increased labor income and an individualized MWP policy could then be considered in this respect. All in all, what can be achieved by family- and individual-based policies is not so clear and requires thorough investigation based on behavioral modeling. In the following, we compare both types of policy measures, choosing the British WFTC as a benchmark for family-based policies.

2.3. Recent trends in the UK and the three countries under study

The WFTC was introduced in the UK in October 1999. It is a transfer to households with children where at least one of the adults is in paid work (employment or self-employment) for at least 16 hours per week; a 20% premium is given if one of the working adult is above 30 hours per week; maximum entitlement also increases by 49% per dependent child. Eligibility is based on jointly assessed income in the household; once income reaches a threshold level, the maximum amount is tapered away at a rate of 55% on net income. The WFTC is a more generous variant of the previous *Family Credit* (FC); for instance, the maximum yearly entitlement for a lone parent with one child was £3,180 in 1998 (FC) compared to £4,160 in 2001 (WFTC).

Recent studies rely mostly on ex-ante evaluations based on tax-benefit calculators and structural models of labor supply.⁵ Blundell et al. (2000) find a relatively satisfying distributional effect of the reform while they predict a mitigate effect on employment. Participation of single women would increase by 34,000 whereas 20,000 married women with employed partners would stop working. When combining other minor changes, the WFTC leads to a small increase

⁴ Note, however, that if the primary policy goal is to reduce poverty, a family-based MWP scheme may not be an appropriate instrument insofar as it is fundamentally a transfer to the 'working poor', who, arguably, are not the poorest in the low income population.

⁵ Blundell et al. (2000) suggest a difference-in-difference analysis as a first check of the ex-ante evaluation. More recently, Francesconi and Van der Klauuw (2004) have studied the effect of the 1999 WFTC on single women using panel data. They find larger responses and a more diverse picture across responses than in ex-ante evaluations.

in overall participation by just above 27,000 individuals. Consequently, the distributive impact of the reform—rather than the incentive effects—has been appealed to justify the large cost of the reform. The UK experience clearly illustrates the difficulty of attaining multiple objectives through a single instrument. As a matter of fact, the new reform of April 2003 has splitted the credit into a refundable Child Tax Credit (CTC) and a Working Tax Credit extended to childless households. While the CTC may well contribute to the reduction of child poverty, it is not sure that the WTC—officially aimed at supporting low earnings and encouraging labor market participation—will do better than the WFTC in achieving both objectives.

Let us now turn to the recent reform agenda in Belgium and in the three countries under study. The 2001 Belgian reform provides an interesting example of an individual refundable tax credit on earned income (*Crédit d'impôt sur les bas revenus de l'activité professionnelle*). In France, a refundable tax credit has also been implemented in 2001 (*Prime pour l'Emploi*, PPE) but is means-tested at both personal and household levels. In both countries, the measures target workers working full time at the minimum wage. However, maximum entitlements (around 500 EUR per year in both countries) are too modest to have significant effects on work incentives (see Orsini, 2004; Bargain, 2004). The Belgian tax credit has recently been replaced by a reduction of employees' social security contributions levied on low earnings. In comparison, neither Finland nor Germany have introduced in work transfers *stricto sensu*. Both countries have extended the level of earned income *allowance* on income taxation. This concerns the flat-rate local taxation in Finland (see Laine, 2002) and the progressive taxation in Germany (see Haan and Steiner, 2004). As opposed to tax credits, tax allowances consist of a fixed deduction from the taxable income; in particular, they are not refundable and the net gain for tax payer is somewhat limited, i.e. around 700 EUR in Finland and 1,115 EUR (in the first tax bracket) in Germany in 2005. Exemptions on social security contributions have also been extended in 2003 in Germany and now apply to gross earnings below 400 EUR, according to the so-called Mini-jobs reform.

Overall, the budget allocated to these recent reforms in Finland, Germany and France is well below the level of transfers implied by the British scheme. An additional motivation for the present study is to question what could have happened, had these countries dedicated the same budgetary expenses as the UK to MWP transfers.

3. Simulation of in-work transfers in three European countries

3.1. Description of the reforms

The reforms simulated hereafter are in line with the two broad groups of policies surveyed in the previous section. We have opted on the one hand for a working tax credit (WTC) conditioned on family income, and, on the other hand, for an individualized low-wage subsidy (LWS).⁶ To target those with a significant degree of participation, the LWS is proportional to work duration while WTC eligibility is conditioned on working at least 16 hours. We presume that reforms are financed in a way which does not affect labor supply behavior or vertical distribution (e.g. an increase in indirect taxation). More importantly, we assure comparability of the reforms by equalizing fiscal costs as described below. We now provide a summary of each reform (more details can be found in Bargain and Orsini, *in press*). Simply note that simulations

⁶ In this paper, we focus on financial incentives only and ignore the type of administrative arrangements, e.g. frequency of payment, identity of the recipient in the household, etc.

continuously try to balance international comparability and overall coherence in each institutional setting.

The **working tax credit** is based on the essential features of the 2001 British WFTC, as described in Section 2, but extended to childless singles and couples.⁷ The formula to compute total household entitlement is as follows:

$$\text{WTC} = B - \max(0; (z - \theta)t)$$

with B the maximum theoretical amount, θ the threshold or disregard, t the taper rate applied to (jointly assessed) net income z . Parameters B and θ are taken from the 2001 WFTC rules and converted to 1998.⁸ The taper rate t equals 55% on net income, which corresponds roughly to a 37% taper on gross income in the UK and to smaller percentages in continental Europe due to higher levels of taxation. The childcare credit and a further condition on wealth are not modeled here. For the 2001 WFTC the assessment of family income z is the same as for social assistance (*Income Support*): all main sources of income are included net of taxes and social security contributions and all family benefits are included, with the exception of the Child Benefit, the Maternity Benefit and the Statutory Maternity Pay (plus small UK-specific disregards). In a similar way, WTC assessment is modeled along the lines of country-specific social assistance benefits. Yet, universal child benefits are excluded from the means-test while usually being part of income assessment for social assistance in France and Finland.⁹

By definition, the **low wage subsidy** is individual and does not account for the family dimension nor for the presence of other incomes. It consists of increasing wage rates w , hence gross labor income $y = wh$, by a percentage A . The LWS decreases if the wage rate is larger than a reference wage W until it falls to zero at an upper bound $1.4 W$.¹⁰ The formula to compute the level of the LWS is then written as follows:

$$\text{LWS} = Ay \text{ if } w \leq W$$

$$\text{LWS} = KAy \text{ if } w \in [W, 1.4W]$$

with $K = ((1.4 - w/W))/(0.4) \in [0, 1]$. There are no official wage floors—a natural choice for W —in Germany and Finland. Instead, we opt for the 10% cut-point of the wage distribution of each country, which corresponds to 6.09 EUR per hour for France (close to the 1998 French minimum

⁷ Notice that it does not correspond to the reform implemented in the UK in 2003. In the new British system, the child premium is universalized in a new instrument (the Child Tax Credit). Instead, we have maintained the child element of the WFTC. In effect, since the policy is aimed at recreating significant financial difference between social transfers and paid work, it must be scaled on family size just as social transfers are.

⁸ Absolute amounts are converted using 1998 exchange rates (0.67833 £/EUR) and an inflation of 6.8% over the 1998–2001 period. This gives $B = 74.6$ EUR/week for a childless household and $\theta = 128.3$ EUR/week. Alternatively, we could have chosen relative amounts computed as a function of a national reference such as the average equivalized income.

⁹ Another sensitive issue in modeling WTC concerns the way it interacts with the rest of the system. As in the UK, WTC is not itself taxable but enters the income assessment of the Income Support (which impacts in turn on the computation of housing benefits in Germany). In Finland and France housing benefits condition the computation of social assistance. For comparability purposes, then, WTC does not impact on housing benefits, contrary to the situation in the UK where the interaction of both instruments reduces the net gain of the tax credit.

¹⁰ This choice has been driven by the fact that the individual tax credits in France and Belgium were defined originally to reach a maximum for a full time worker paid at the minimum wage and to exhaust at 1.4 times the minimum wage. This way, the LWS is by construction a little more tailored to national conditions than the WTC, since eligibility to the former is linked to the wage distribution while in the latter, absolute criteria are retained. Optimally, any reform, whether family or individual based, should be designed to adapt to the national wage/income distribution as detailed below.

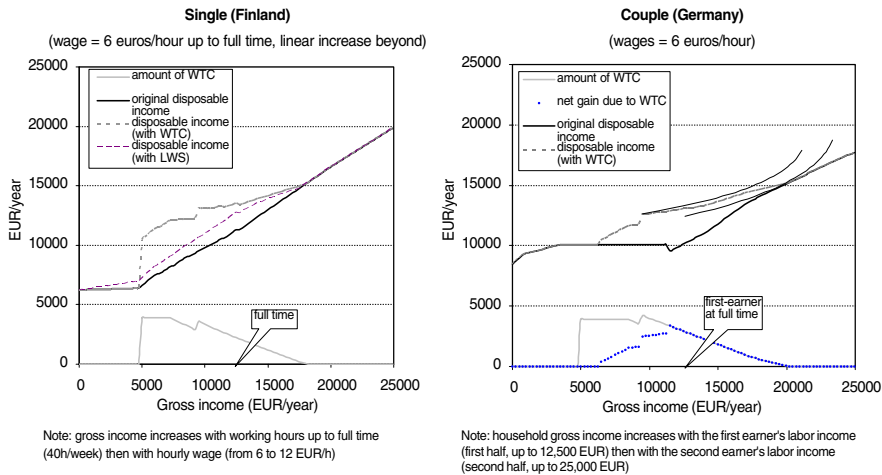


Fig. 2. Impacts of the reforms on budget curves.

wage), 6.79 for Finland and 7.42 for Germany. An important aspect of this study is that for each country, we calibrate factor A in order to reach the same *real* cost, i.e. after labor supply adjustments, for both reforms. We find $A=12\%$ for Finland, 20.5% for France and 13% for Germany.

3.2. Impact of the reforms on budget curves

We now look at hypothetical budget constraints in Fig. 2 to comment the effects of the reform on the systems in force. It must be stressed that these examples are merely illustrative and shall not be generalized too widely.

The l.h.s. graph depicts the budget constraint of a single individual in activity (here for Finland). The vertical axis represents disposable income in function of gross earnings on the horizontal axis. For illustration, assume that the first half of the horizontal axis corresponds to a linear increase in weekly hours of work from 0 to 40 (while wage rate equals 6 EUR per hour), while the second half corresponds to a linear increase in the hourly wage from 6 to 12 EUR. With this definition, the amount of wage supplement A_{wh} first increases linearly with working time at a flat rate A_w so that the slope of the budget curve gets steeper. After hitting 40 hours, the wage rate increases and the benefit starts to decrease as soon as it exceeds the reference wage (6.79 EUR for Finland). There is actually no phasing-out on earnings with this reform (the LWS is conditioned on wage rates) so that the LWS must decrease EMTRs of all eligible individuals. It also contributes to a modest increase in financial gains to work.

The amount of WTC is depicted at the bottom of the graph and the threshold at 16 hours—marking the eligibility—is particularly evident, as well as the discontinuity for the 30 hours premium. Looking at the budget constraints, it appears that the WTC creates large gains to work. Yet, the WTC flattens the budget curve in a way which dramatically increases EMTRs for individuals in the phase-out range.¹¹ For individuals in that range and beyond it may be optimal to decrease working hours and switch, for instance, from full- to part-time activity. Overall, the

¹¹ Only in Germany, where social assistance and WTC interact, some of the persons on welfare benefit from the 55% taper rate instead of the 100% withdrawal rate of the *Sozialhilfe*.

WTC unequivocally enhances the probability of participation for single individuals or lone parents but the impact on working hours is more ambiguous.

The r.h.s. graph in Fig. 2 illustrates the budget constraint for a couple (here for Germany). The first half of the horizontal axis corresponds to a linear increase in weekly hours of work from 0 to 40 for the main earner while the second half represents a linear increase in weekly hours of work from 0 to 40 for the secondary earner. The difference between the pre-reform disposable income and the disposable income with WTC is the net gain from the WTC, depicted at the bottom of the graph (this is simply the difference between the two budget curves). This gain turns out to be much smaller than the amount of transfer, due to the fact that the WTC enters the income assessment of social assistance. This effect occurs in all countries and is maximal in the case of Germany, due to a more generous safety net.¹² In this case, the net gains reach a maximum in a range between 10,000 and 15,000 EUR instead of the 5,000–10,000 range initially targeted by the reform.¹³ Nonetheless, in all countries the WTC creates a significant financial difference between non-participation and full-time activity for the main earner. As earnings of the secondary earner increase, however, the amount of WTC received through the pay of the main earner is phased out. The reform then typically reduces net gains for wives and induces them to move out of work. This is illustrated by a shift of the indifference curve depicted in the graph, primarily tangent to the pre-reform budget constraint at a point where the wife is in activity, then tangent at a point where she is inactive and where the husband is at the 30 hours bonus peak.

4. The behavioral model

4.1. *Microsimulation, data and sample selection*

The simulation of the actual tax-benefit systems and the reforms are conducted using EUROMOD, an integrated tax and benefit calculator for the EU-15 countries. For each country and for the year 1998, the program covers the computation of all social contributions, direct taxes and transfers (including child benefit, housing benefit and social assistance) to yield household disposable income. EUROMOD is based on micro-data which have been rendered homogeneous across countries, including similar variables definitions (see [Sutherland, 2001](#), for a comprehensive description). Finnish data are selected from the 1998. Income Distribution Survey which contains socio-economic information on 9,345 representative households. German data come from the 1998 wave of the German Socio-Economic Panel (GSOEP) which contains information on 7,677 representative households. The data used for France are taken from the 1994 Household Budget Survey which contains information on 11,220 representative households; monetary variables have been inflated to year 1998; over the 1994–1998 period, demographics are assumed constant and no structural change occurs in the tax-benefit system.

The reforms are simulated for each country on the complete representative sample while labor supply is estimated on selected subgroups of couples and single women, as explained below. For that purpose we only keep households where adults are aged between 25 and 64

¹² Working 16 hours at minimum wage in France and in Finland is sufficient to exit this segment of income assistance while the same does not hold for Germany.

¹³ This effect did not appear in the previous example on a Finnish single individual (l.h.s. graph) as his eligibility ceases approximately at the point where eligibility to WTC begins (16 hours). Also note that the LWS, not represented here, does not interact at all with the existing system.

and available for the labor market, i.e. neither disabled nor retired nor in education. Self-employed or farmers are excluded as their labor supply decisions are probably very different from those of salary workers. To further increase data homogeneity, ‘extreme’ households are selected out, i.e. households with more than 3 children; who receive important levels of non-labor income; whose children earn substantial earnings.

A difficult issue concerns the treatment of unemployed workers as opposed to individuals receiving social assistance. Most often, in the absence of modeling of the demand-side, the former are treated as other inactive workers, which amounts to arguing that they voluntarily remain unemployed. This is an extreme hypothesis, especially in the context of continental European labor markets. On the other hand, assuming like here that all unemployed persons are rationed is also arbitrary. We simply argue that including them would require additional information and a specific modeling of the employment process (see Laroque and Salanié, 2002), much like the withdrawal of self-employed workers. More fundamentally, this choice conforms to the presumption that social assistance discourages labor market participation so that primary targets of in-work benefits are specifically households on welfare, and not those receiving unemployment benefits.¹⁴ Descriptive statistics of the selected samples are presented in Table 2.

4.2. A structural model of labor supply

We use a discrete-choice model of labor supply that is uniformly applied to the three countries under study. Similarly to Blundell et al. (2000) for the evaluation of the WFTC, we make use of an explicit parameterization of consumption-leisure preferences. The problem is reduced to utility-maximization among a discrete set of hours, which allows to handle non-linear and non-convex budget sets—due to taxes and means-tested benefits—in a straightforward way.

As male labor supply displays less variability than female working hours, it is often reduced to the modeling of participation decisions only. Yet, male participation rates are extremely high in the selected samples, especially for France and Germany (see Table 2); attempts to estimate a joint labor supply model with binary choices for men did not lead to robust results. Consequently, we simply treat male labor supply as fixed at observed values.¹⁵

¹⁴ The supply-side treatment for the unemployed group relies on other types of policy, in particular reductions in the generosity and duration of unemployment insurance (see Pellizzari, 2004, for an overview and on the combined effects of unemployment insurance and social assistance). On average, this group has probably higher reservation wages and is less concerned by the incremental incentive provided by in-work policies. An intermediary category of people – those receiving low levels of unemployment benefit or having exhausted their right – are kept in the selection and treated as if on welfare (for instance, the recipients of ‘Unemployment Assistance’ in Germany or of the ‘Allocation Spécifique de Solidarité’ in France).

¹⁵ Male labor supply is known to be inelastic to exogenous changes of moderate size in the budget constraint. This has been proven extensively in numerous studies, as surveyed by Pencavel (1986) or Blundell and MaCurdy (2000); a convincing and recent proof is given by Eissa and Hoynes (2004) who show that the large expansion of the EITC in the US induces extremely small responses from married men. Ignoring male responses here does not seem a large understatement for married men – most of whom work. Things are more problematic for single men, who represent a substantial proportion of households on welfare (from a fifth in Germany to a third in France); those may be encouraged to take up a job with the reforms. However, single men present two specific problems: (1) those receiving SA are also often working in the parallel economy, so that our models here do not apply without a better knowledge of their true reservation wage, (2) ‘inactive’ single men form the category typically under-represented in most household surveys; in the present case, only very few of them remain after applying the selection process. To capture the impact of in-work policies on this sub-population, specific datasets are required, as the survey on the French minimum income recipients used by Gurgand and Margolis (2001).

Table 2
Descriptive statistics

	France			Germany			Finland		
	Women in couple	Men in couple	Single women	Women in couple	Men in couple	Single women	Women in couple	Men in couple	Single women
Participation*	70.5%	99.1%	96.8%	63.2%	95.7%	79.3%	73.6%	90.1%	80.1%
Working time (hours/week)/participants	35.7	42.0	37.4	33.3	38.2	36.0	37.0	40.0	37.7
Working time (hours/week)/all	26.0	41.9	36.6	21.6	36.7	28.7	35.5	39.8	36.7
Gross wage rate (euros/hour)/participants	10.5	12.8	11.2	12.2	15.8	12.8	11.8	15.5	12.5
Gross wage rate (euros/hour)/out-of-work**	8.7	n.a.	9.9	11.1	n.a.	11.2	10.2	n.a.	9.0
Average age	38.2	40.4	40.6	38.4	40.9	38.5	40.2	41.8	42.5
Primary education	30.7%	17.9%	22.3%	14.6%	11.1%	21.0%	16.1%	18.6%	17.6%
Vocational training	37.9%	46.0%	32.7%	48.7%	44.8%	44.6%	35.2%	37.8%	36.1%
High school diploma	14.8%	17.9%	15.1%	23.2%	25.9%	22.5%	27.5%	18.1%	26.2%
University studies	16.7%	18.2%	30.0%	8.5%	14.9%	11.9%	21.2%	25.4%	20.2%
Average number of children	1.47		0.64	1.11		0.59	1.19		0.54
Presence of child 0–2	17.1%		2.3%	11.9%		4.9%	15.9%		1.9%
Presence of child 3–5	20.2%		4.8%	15.7%		6.2%	18.6%		5.3%
Presence of child 6–11	33.8%		14.5%	29.4%		19.2%	29.4%		12.0%
No. of selected households	2,095		664	1,265		453	1,632		416
Corresponding population	3,898,106		1,458,464	6,285,220		2,250,754	329,343		171,100
% of total population	16.9%		6.3%	16.5%		5.9%	14.0%		7.3%

* Definition of participation as retained in our discretization (i.e working at least 15 hours per week).

** Wage prediction.

The choice to focus on female labor supply is well documented in the literature (see the discussion in [Laroque and Salanié, 2002](#)), and women form the group with the highest margins for increasing labor supply. Let us distinguish between married and single women. *Married women* present the lowest participation rates due to a range of factors: social norms, gender wage gaps, institutional constraints responsible for large fixed costs or lower implicit wages due to joint taxation (in France and Germany), rationing in childcare facilities (in Germany). *Single women* and especially lone mothers represent a group of particular interest since their gain to work is amongst the lowest, due to the generosity of social transfers in presence of children. Even so, many of them work for small part-time or temporary jobs and cumulate assistance and low earnings. In France and Germany, around 45% of the social assistance recipients are single women, more than half of whom are mothers; in Germany, one quarter of all single mothers are on welfare.

The distributions of actual hours for the selected samples are represented in [Fig. 3](#). The pattern of hours presents strong concentrations around full time activity and small concentration on part-time in some cases. This is not surprising since continental Europe is characterized by strong demand-side and institutional rigidities which constrain salaried workers to choose from a limited set of options. A discrete-choice model of labor supply seems fairly appropriate in these circumstances (see [Van Soest, 1995](#)); to ease comparison across countries, we suggest a simple discretization, allowing non-participation, part-time and full-time work. This choice also conforms to the well known fact that responsiveness is mainly driven by variations at the extensive margin (see [Heckman, 1993](#)).¹⁶

We rely hereafter on the conditional logit model. If household i is offered to choose one from J possible work durations for the female adult, it is assumed that the utility the household may derive from alternative j ($= 1, \dots, J$) is given by:

$$V_{ij} = U(H_j, C_{ij}, Z_i) + \epsilon_{ij},$$

where U is a conventional utility function which depends on female working duration (H_j) and consumption (C_{ij}) as well as on a vector Z_i of household socio-demographic characteristics. Women are assumed to choose between $H_1=0$, $H_2=20$ and $H_3=39$ hours per week. The actual utility derived from alternative j for household i , V_{ij} , also includes an error term ϵ_{ij} , assumed to be identically and independently distributed across alternatives and across households according to a EV-I distribution. Under this assumption, [McFadden \(1973\)](#) proves that the probability that alternative k is chosen by household i is given by:

$$P_{ik} = \Pr(V_{ik} \geq V_{ij}, \forall j = 1, \dots, J) = \frac{\exp U(H_k, C_{ik}, Z_i)}{\sum_{j=1}^J \exp U(H_j, C_{ij}, Z_i)}. \quad (1)$$

We choose a quadratic specification of the utility function as in [Blundell et al. \(2000\)](#):

$$U_{ij} = \alpha_{cc}(C_{ij} - F_{ij})^2 + \alpha_{hh}(H_j)^2 + \alpha_{ch}(C_{ij} - F_{ij})H_j + \alpha_{ci}(C_{ij} - F_{ij}) + \alpha_{hi}H_j, \quad (2)$$

¹⁶ Interestingly, recent proofs are precisely provided by natural experiments related to in-work transfers. In particular, [Meyer \(2003\)](#) studies changes in the EITC between 1990 and 1996 in the US and shows that nearly all of the labor supply adjustment by single mothers in response to these changes occurred at the extensive margin. The weakness of adjustment in working hours is merely explained by behavior and not connected to an explanation in terms of rationing of worked hours, as the US labor market presents a much more continuous distribution of work hours than continental Europe.

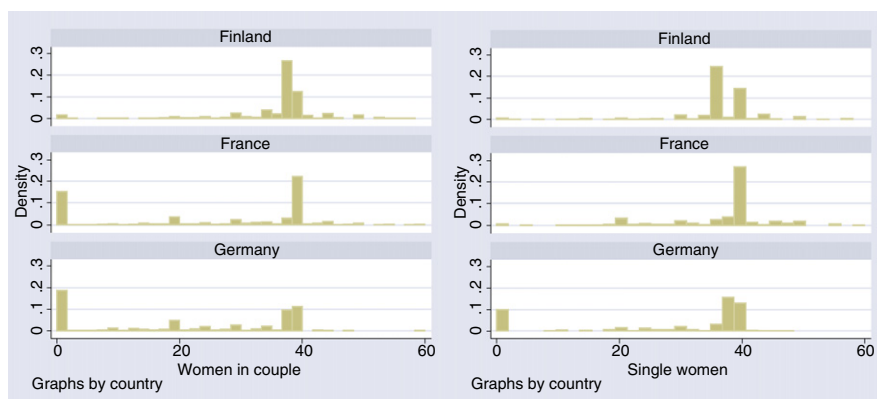


Fig. 3. Distribution of working time (hours/week).

with observed heterogeneity:

$$\alpha_{ci} = \alpha_c^0 + \alpha_c' Z_i$$

$$\alpha_{hi} = \alpha_h^0 + \alpha_h' Z_i,$$

Vect or Z_i contains usual taste shifters as age, number of children in various age brackets and a dummy for living in the main city. Costs of work F_{ij} are also taken out of disposable income. They vary linearly with educational dummies and with the number of very young children (supposed to represent potential child care costs when the wife is working).

In the present static framework, consumption C_{ij} is equivalent to disposable income $D(w_i H_j, y_i, Z_i)$ which itself is expressed as a function of household characteristics, exogenous income y_i (including female unearned income and the earnings of the husband for women in couples) and gross labor income of the wife $w_i H_j$ (for any alternative j). Hourly gross wage rates w_i are assumed not to vary with work hours; they are computed from the data or predicted for non-participants.¹⁷ Function D relies on a fairly complex set of tax-benefit rules, computed for each discrete working hour using EUROMOD.

The likelihood of a sample of observed choices can be derived from expression 1 as a function of the preference parameters α whose estimates are obtained by the ML technique. Among usual regularity conditions, positive monotonicity and quasi-concavity in consumption seem natural minimum requirements and are imposed as constraints within the likelihood maximization. Note that the model is unidentified non-parametrically, as in related studies. In particular, information on children enters preferences, fixed cost and disposable income through the tax-benefit rules. Structural identification is allowed by the specific functional form retained here (interaction terms of children variable with C and H) and the nonlinearity of the tax-benefit rules.

4.3. Results of the estimations

The present approach follows the bulk of the literature and does not account explicitly for the demand-side, for life-cycle aspects nor for domestic production, so that one might be cautious

¹⁷ We use the usual estimation technique with Heckman correction of the selection bias; identification of the participation probit relies on female unearned incomes and the presence of young children (results available upon request). As reported in Table 2, wage rates for non-participants are significantly lower than for participants.

when interpreting the results purely in terms of preferences in a static environment. Let us detail some of these issues. Firstly, rationing in hours or in participation may significantly influence the results. As we shall see below, fixed costs of work and educational dummies might capture part of it. Secondly, a few low-wage individuals are observed working while their income is topped up by social assistance. The problem seems to occur essentially for French single women working part-time, 14% of whom have effectively no financial return from doing so. They do not behave according to the static rationality presumed by the model. There may be a long run pay-off to labor market attachment for these workers, particularly through wage progression, that our static modeling is indeed unable to explain. To some extent, this implies that we should capture lower distaste for work or smaller taste for income than is actually the case. All in all, it is very unlikely that ‘consumption-leisure preferences’ captured in the model display regular properties, in particular with respect to hours of work. Thereby, we do not impose regularity conditions on hours. We also refrain from using the model for welfare analysis due to previous arguments on the lack of robustness of the model interpretation.

Notwithstanding previous restrictions, we compare labor supply choices in France, Germany and Finland with respect to the role of household characteristics. [Tables 8 and 9](#) in the Appendices present the results of the estimations for women in couples and single women respectively. In both cases, very few taste parameters for income are significant. Things look better for hours; as expected, the utility of work decreases with the presence of children, especially very young children. Women prefer to work significantly more if located in East Germany, while living in Paris or Helsinki does not seem to alter behaviors in a significant way. The impact of age displays the usual parabolic pattern—except for German women in couples and for French singles where it is not significant—suggesting a move towards single-earner couples as the household ages or a cohort effect.

The average fixed cost of work is significant only in Germany. For married women, fixed costs associated with young children are significant in France and Germany; larger in Germany, they could reflect important differences in access to childcare facilities between these countries. The dummy for low (primary) education turns out to have a significant negative effect on labor supply in France, which could translate that rationing primarily concerns less educated people (e.g. additional disutility from job search), or relate to preferences (more educated ones are more hard-working) or psychological costs to work (women with higher education levels may find it more rewarding to take a job since it is likely to be more attractive). For single women, fixed costs are not included for Finland since there are very few part-timers.

Goodness-of-fit in estimation tables is expressed in terms of the pseudo-R². This indicator is helpful for the specification but is not informative in absolute term to compare fits across countries. Additionally, [Table 10](#) in the Appendices shows that predicted probabilities for each choice correctly represent the proportions of the samples for all countries. The generalized R², i.e. the percentage of observed variance explained by the model, shows a very poor prediction of part-time work, as in previous related studies ([Laroque and Salanié, 2002](#), for France; [Bonin et al., 2002](#), for Germany).

5. Framework conditions and the design of tax-benefit policies

As noted in the introduction, initial or ‘framework conditions’ give important insights to understand how given policies may achieve their objectives. These conditions should deserve special attention during the process of designing a policy in relation to country-specific circumstances. In the present exercise, they are analyzed ex-post, since the reforms are already

Table 3
Distribution of households according to income ranges in WTC rules

	France		Germany		Finland	
	Freq.	Percent (%)	Freq.	Percent (%)	Freq.	Percent (%)
Not eligible	4,867,352	29.8	18,755,686	53.6	566,249	33.5
Potentially eligible*	11,444,148	70.2	16,236,265	46.4	1,123,038	66.5
– at maximum benefit	172,902	1.5	87,480	0.5	7,095	0.6
– in the phase-out range	2,143,331	18.7	3,355,728	20.7	241,568	21.5
– over the benefit range	9,127,915	79.8	12,793,057	78.8	874,375	77.9

Source: authors' computations using Euromod.

* Households with at least one adult working at least 16 hours per week.

defined); cross-country heterogeneity explains the primary differences in the effects of the reforms.

5.1. Participation rate and distribution of earnings/wage rates

Participation rates give a first indication on the proportion of households potentially eligible for a transfer conditional to work. More precisely, in the case of the WTC, we must take a look at the proportion of households with at least one member working a minimum of 16 hours per week. Table 3 shows the proportion of these 'potentially eligible' households, in percentage of the total number of households with head or spouse in working age. Accordingly, while eligibility for the WTC also depends on household income, the population of recipients should be somewhat lower in Germany.

Fig. 4 represent both the distributions of wage rates and labor incomes (in proportion of the sample mean) for female workers.¹⁸ These distributions determine to some extent the targeting of the reforms. The graphs reveal that the female wage structure is more compressed in Finland than in France and in France more so than in Germany, an observation in line with the literature on wage inequalities. The presence of a wage floor in France induces important asymmetries in the distributions—especially in the wage rate—compared to the other countries.

The *distribution of incomes* is obviously what matters for the design of the WTC. The high percentage of females working at minimum wage explains that the largest proportion of recipients in the maximum benefit range (i.e. the flat segment) is found in France, while the largest proportion in the phase-out range is found in Finland, according to Table 3. Even though differences are not very marked, the combined effects of participation levels and earnings distribution must ensure a higher proportion of recipients in France and Finland compared to Germany. Overall, the extremely small number of households in the maximum benefit range is striking. It is actually explained by the fact that full benefit requires net earnings to stand below $\theta = 128.3$ EUR per week, which corresponds to some rare cases of single-earner households with the working person employed part-time for a maximum of 6.4 EUR/hour.

In the case of the LWS policy only the differences in *wage rate distributions* must be considered. It is important to remember that the eligibility region is tailored to the wage distribution, i.e. LWS is distributed fully up to the country-specific reference wage W and decreases with the wage rate to exhaust at $1.4 W$. Consequently, the concentration of wage rates in the W – $1.4 W$ region conditions the number of eligible individuals. Here too, the modes in

¹⁸ Similar graphs for men are not represented as differences in concentrations across countries are less significant. They are available upon request.

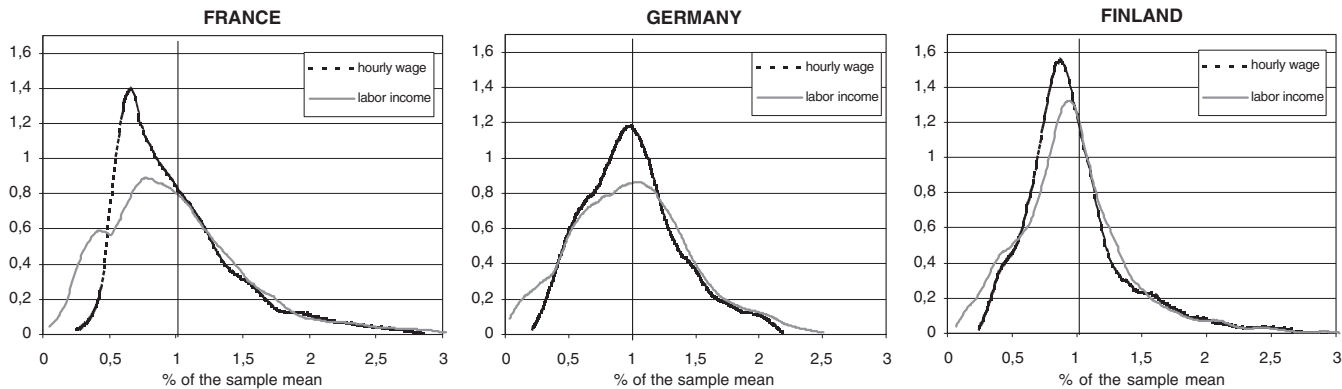


Fig. 4. Distribution of hourly wage rate and labor income for active women.

France and Finland are located in lower ranges of values, while higher concentration in Finland could imply relatively more eligible individuals, *ceteris paribus*.

5.2. Potential effects on hours and participation

A crucial issue concerns the way a reform interacts with the existing system. This aspect of particular importance in the case of the WTC has been studied briefly in the analysis of budget constraints. The latter cannot be generalized, however, and a comprehensive picture of the impact of the reforms is obtained by looking at the change in the distribution of EMTRs and financial gains to work.

Variations in EMTRs imply potential effects at the intensive margin and, in our discrete framework, translate into shifts between part-time and full-time. A detailed analysis is conducted in Bargain and Orsini (in press) and we provide here a short summary of the results. With the introduction of the WTC, EMTRs increase substantially for deciles 3 to 5, given the large phasing-out of the transfer. Despite a uniform taper rate on net income, the rise in EMTRs is effectively more important in France where taxation is relatively lower. Moves from full-time to part-time are to be expected. In contrast, the LWS reform shifts EMTRs downwards along the whole income distribution, while strongest reductions occur between deciles 2 to 4.

Variations in financial gains to work indicate potential changes in participation. The gains are simply computed as the relative increase in household disposable income when the woman switches from inactivity to full- or part-time work. Larger amounts of transfer through a well targeted WTC naturally lead to a larger increase in financial gains to work for *single women*. This is particularly striking with regard to the gain of working *part-time*, which rises from 47% to 79% in France, from 67% to 91% in Germany and from 56% to 81% in Finland. The increase is much smaller with the LWS (only 2 or 3 percentage points). The average gain of working *full-time* increases by 7 percentage points in Finland and by more than 10 points in Germany and France with the WTC; it increases only by half of this when the LWS is introduced. Similarly, the proportion of very low gains (less than 40%) decreases substantially in all three countries when the WTC is introduced.

However, for *married women* the picture is completely different. Firstly, it is expected that in the pre-reform situation relative gains are much smaller for married than for single women; this is especially the case in France and Germany where the earnings of the secondary earner are taxed away at the marginal tax rate of the first earner, a consequence of the joint income taxation system. As a matter of fact, the same effect applies with the WTC which is conditional on family income. As a result, additional earnings by wives may lead to a loss in WTC entitlement for their working partners, as seen in the budget curve analysis. The gain of working full-time thus shifts from 58% to 50% in France, from 58% to 53% in Germany and from 65% to 61% in Finland. In addition, the proportion of small gains associated to working full-time (less than 30%) increases drastically in France (from 9.5% to 20% of the selected women) and more moderately in Germany and Finland (from 17% to 23% and from 7.7% to 10.8% respectively). The LWS slightly improves average gains of working (especially full-time) and reduces the proportion of small gains, especially in France and Germany.

5.3. Labor supply elasticities

A key issue in determining the impact of reforms is the elasticity of labor supply to exogenous changes in budget constraints. Using previous estimates, we simply compute own wage elasticities

to provide an order of magnitude of the labor supply responsiveness across countries. Evaluating elasticities at the sample mean is not very informative—in a highly nonlinear model like ours—about the consequence of wage changes in a heterogeneous population. Instead, we compute wage elasticities numerically and average over the whole sample. To do so, we increase female wage rates uniformly by 1% (alternatively by 10% for a check) and simulate in each case the subsequent changes in average work duration and in the participation rate.¹⁹ Importantly, note that elasticities of working hours account by construction for participation effects.

Predicted elasticities presented in Table 4 are overall in line with the recent labor supply literature (see Blundell and MaCurdy, 2000). Significant differences across country are not surprising and reflect principally differences in preferences, fixed costs of work and tax and benefit legislation. For married women, rationing on childcare facilities in Germany translates into larger fixed costs, hence larger reservation wages; in consequence, marginal increases in wage rates may not lead to a switch from inactivity to participation as often as in France, hence smaller elasticities. The point is of lesser importance regarding single women, for whom the ranking is reversed; elasticities are particularly small in France, where the level of participation is much higher than in Germany and Finland. Note that the figures presented here are only average values while it may be interesting to compare their distributions.

Comparison with other findings is only indicative as many aspects differ from one study to another. The range of techniques, data selection and even measures of elasticities is large. Nevertheless, the values we obtain for France compare well to the order of magnitude of recent findings, even if slightly higher. Choné et al. (2003) focus on couples with at least one child under the age of seven and find elasticities of participation around 0.3 (when minimum wage is increased simultaneously to avoid the censorship effect). In recent estimations on couples, assuming male labor supply constrained, Donni and Moreau (2005) find an elasticity of 0.47 for women at the mean of the sample.

In Germany, results are in the range of recent estimates provided by Bonin et al. (2002) and Haan and Steiner (2004). Both studies rely on the GSOEP (2000 and 2001 waves respectively). Former authors find own wage-elasticities of 0.27 with respect to working hours in the case of women in couples. Closer to our findings Haan and Steiner (2004) report elasticities of hours of around 0.39 for women in couples (when husbands' labor supply is fixed) and of 0.13 for single women. As in all related studies on Germany, elasticities are found markedly smaller for East German women.

Smaller wage-elasticities for married women in Finland coincide with a higher proportion of women working full-time in this country. Yet, there are few studies related to labor supply estimations for Finland. For the year 1987 (before the Finnish recession), Ilmakunnas (1992) considers only working women in couples and finds uncompensated wage elasticities in a range between 0.09 and 0.11. Kuusmanen (2000) finds very small responses to important changes in the tax system and Laine (2002) provides difference-in-difference estimations of the impact of the 1996–2001 reforms, finding very moderate effects. Overall, it seems that responsiveness is extremely small in Finland. Although robust to several specification of the labor supply model,

¹⁹ Transition frequencies are the means over 200 simulated transitions. Each transition is obtained by calibration of the stochastic part of the utility at each hour choice in order to obtain a perfect match between observed and predicted hours for the pre-reform situation. Confidence intervals for each transition cell and summary measure are simulated by drawing 200 times from the estimated asymptotic distribution of the parameter estimates and for each of those parameter draws, applying the calibration method to build transition matrices. The same technique is applied to simulate the labor supply effects of a reform.

Table 4
Elasticities of female labor supply

Country	Women in couples		Single women	
	Hours	Participation	Hours	Participation
France	0.59 [0.52; 0.65]	0.52 [0.46; 0.58]	0.11 [0.08; 0.14]	0.06 [0.04; 0.07]
Finland	0.14 [0.10; 0.18]	0.14 [0.10; 0.17]	0.27 [0.18; 0.34]	0.26 [0.18; 0.33]
Germany	0.38 [0.31; 0.45]	0.32 [0.27; 0.38]	0.14 [0.09; 0.18]	0.12 [0.08; 0.15]

Elasticities are computed using averaged simulated transitions; figures in brackets give bootstrapped 90% confidence intervals.

our results show larger elasticities for single than for married women, which seems surprising and requires further investigation.

6. A behavioral analysis of the reforms

6.1. Labor supply responses

Labor supply responses to the WTC are presented in the upper part of Table 5. Around 1.8% of the sample of single women in Germany and in Finland are encouraged to enter the labor market while the effect is small in France (0.51%). Even though the increase in the financial gains is of comparable magnitude across countries, results turn out to be much larger in these two countries since (i) the pre-reform participation rate is comparatively much lower in Germany than in France and (ii) participation elasticities for singles are substantially larger in Finland than in other countries. This positive participation effect essentially concerns lone mothers (around 80% of the movers in all countries). Transitions from full- to part-time activity due to increased EMTRs are especially numerous in France (6.17% of the selected single women).

As for *women in couples*, simulations confirm the analysis on financial gain to work. The WTC introduces a strong bias towards the participation of married women whose partners work: 4.32%, 1.43% and 1.17% of our selection of women in couples would leave their job in France, Germany and Finland respectively.²⁰ In all countries, almost all negative responses concern women with children (97% in France, 100% in Germany) while the small number of positive responses in Germany (0.35% of the selected women in couples) is essentially due to childless couples. The negative participation effect is especially large in France as a result of two factors: gains of working decrease dramatically in this country and labor supply elasticity of married women is the highest. Note also that married women with children—the group concerned by negative responses—has larger proportion in France (80% of the selected couples) than in Germany (66%) and Finland (61%).

Overall, the disincentive effect for married women prevails so that the *net effect on employment is negative* in all three countries and proportionally larger in France (a net 3% of the selected population would withdraw from the labor market) than in Germany and Finland (respectively 0.60 and 0.14%). Confidence intervals displayed in Table 5 are small enough to confirm the robustness of these results.

²⁰ Note that the associated earnings loss is partially compensated for by an increased tax credit on the husband's earnings while additional utility is theoretically drawn from more leisure (or domestic production, not modeled as such in the present setting).

Table 5
Responses to the Reforms

Reform	Country	Type	Simulated responses to WTC (%)								In % of the selected population
			Non-work to work	Work to non-work (%)	Part-time to full-time (%)	Full-time to part-time (%)	Net effect on employment (%)	Average number of hh	90% confidence intervals		
WTC	France	Married women	0.03	4.35	0.00	0.07	−4.32	−168,405	−187,933	−147,794	−3.14
		Single women	0.51	0.00	0.01	6.17	0.51	7,468	4,773	10,109	0.14
		Total						−160,937	−177,824	−143,021	−3.00
	Germany	Married women	0.35	1.79	0.02	0.14	−1.43	−89,992	−107,122	−71,646	−1.05
		Single women	1.79	0.07	0.30	1.49	1.72	38,708	25,040	53,361	0.45
		Total						−51,284	−53,761	−46,606	−0.60
	Finland	Married women	0.17	1.34	0.00	0.04	−1.17	−3,846	−4,741	−3,267	−0.77
		Single women	1.85	0.00	0.00	0.45	1.85	3,159	2,069	4,562	0.63
		Total						−687	−1,198	−179	−0.14
LWS	France	Married women	3.15	0.05	0.52	0.01	3.10	120,704	106,540	135,977	2.25
		Single women	0.33	0.00	0.64	0.00	0.33	4,865	3,013	6,787	0.09
		Total						125,569	109,553	142,764	2.34
	Germany	Married women	1.02	0.03	0.17	0.01	0.99	62,422	50,479	76,516	0.73
		Single women	0.80	0.00	0.15	0.00	0.80	18,055	11,824	24,643	0.21
		Total						80,477	62,303	101,159	0.94
	Finland	Married women	0.41	0.07	0.02	0.01	0.34	1,115	862	1,329	0.22
		Single women	1.17	0.00	0.01	0.00	1.17	2,004	1,406	2,746	0.40
		Total						3,119	2,268	4,075	0.62

All percentages computed as a proportion of the specific sub-group (singles, couples) except the last column where percentages correspond to the whole selected population.

In contrast, similar reforms in the US and the UK result in a positive effect on singles which offset the negative effect on married women.²¹ Before trying to explain such differences between the two groups of countries, some remarks are required. Firstly, note that in early British estimations, using the same methodology, the net positive effect is too small to draw clear conclusions on the possibility for the WFTC to create work incentives in the UK. Secondly, there is some heterogeneity among the countries in our study and significant differences appear mostly when comparing to France. For married women, Eissa and Hoynes (2004) find that 2.4% of the sample of married women would stop working, which is actually smaller than our findings for France (4.32%) but larger than for Germany (1.43%). Positive effect on single women are quite comparable between Germany (+38,700) and the UK (+34,000 according to Blundell et al., 2000). Thirdly, comparisons are rendered difficult by the fact that the EITC has been extended over a long period (with substantial boosts in the 90s) and that the WFTC is a simple extension of the previous Family Credit; a useful exercise for comparison purposes would be to simulate the complete withdrawal of these instruments. Finally, differences do not come from the fact that we extend the reform to childless couples, as results show that movers are essentially women with children. This said, we can now suggest some interpretations of the differences.

The negative effect on married women is stronger in continental Europe (esp. in France) than in the UK and the US for three reasons. First, the WTC comes in addition to the income splitting system in France and Germany, leading to truly confiscatory implicit marginal tax rates for secondary earners. Second, a smaller share of the population is located in the phase-out range in the US and the UK. In effect, these countries have (i) a wider income distributions and (ii) a lower level of taxation (recall that lower taxation implies a larger taper rate on gross income hence a smaller phase-out region). This explanation is confirmed by Bassanini et al. (1999).²² Third, elasticities of participation are around twice larger in France compared to Germany, the UK and the US (Eissa and Hoynes find an elasticity of 0.267 for married women). At the same time, positive effects on lone mothers are particularly modest in France. According to Christopher et al. (2000), the poverty rate of this group is much lower in France (24.7%) than in Germany (39.1%), the UK (31.9%) and especially the US (47.1%). Mechanically, the WTC thus affects a smaller portion of lone mothers in France. The large poverty rates of lone mothers in Germany explain that those women are well targeted by the WTC and substantially respond to the reform. More fundamentally, the employment level of lone mothers is higher in continental Europe, leaving fewer margins than in the UK for positive responses; it is even higher in our selected sample where (demand-side constrained) job seekers have been withdrawn. This is especially the case in France (employment rate of 95%) and less so in Germany (61%) and Finland (77%); in contrast, only 40% of lone mothers participate before the WFTC reform in the selection used by Blundell et al. (2000).

Labor supply responses to the LWS are presented in the bottom part of Table 5. In contrast to the WTC, the LWS consists of a smaller transfer distributed to a larger number of households. Accordingly, the positive effect on the participation of *single women* is

²¹ Comparison with the US is relatively difficult since studies there either treat single women (ex: Meyer, 2003) or married women (ex: Eissa and Hoynes, 2004) but rarely both at once. British studies more often rely on a comprehensive structural approach which facilitate comparisons with our result. In addition, our simulation is very close to the British WFTC and the institutional setting in Germany, France and Finland is closer to the British than to the US one.

²² The authors simulate a stylized EITC in Germany, Sweden, the UK and the US. Their results suggest that the EITC has positive effect on labor supply in countries with wide income distribution and low income tax (the UK and the US) while the reform could be costly with modest or negative effects for Germany and Sweden.

lower, between half and two-third of what is found with the WTC reform (e.g. 18,000 additional participations versus 39,000 with the WTC in Germany). For *married women*, the LWS reduces the proportion of very small gains to work. Combined with larger elasticities in France and to a lesser extent in Germany, this explains the positive effects on the participation of married women (3.15% of selected women living in couples in France and 1% in Germany).

Overall, the joint positive effect on single and married women leads to the clear conclusion that the LWS could significantly improve social inclusion by enhancing employment in France and, to a lesser extent, in Germany. This result validates the choice made in 2001 by the French government to opt for an individualized policy (see Section 2), even though the actual amounts distributed through the *Prime pour l'emploi* are much smaller than the individual subsidy suggested here. For Finland, Böckerman and Jäntti (2004) stress that demand-side aspects may offer a broader margin of action. As a matter of fact, the Finnish authorities are currently considering possible reductions in employers' social security contributions for low-wage jobs.

6.2. Cost analysis

Table 6 details the cost and targeting of each reform before and after behavioral responses. As mentioned before, some parameters of the LWS are calibrated in order to equalize real costs between the two reforms, i.e. costs after labor supply adjustments (4th and 10th lines in Table 6). Since the WTC has net negative effects on labor supply, this real cost is higher than the pre-response cost. This is due both to the efficiency loss (women stop working) and the fact that husbands start receiving the WTC. Symmetrically, efficiency gains of the LWS guarantee that its real cost is lower than the pre-response cost. Consequently, to equalize real costs between reforms, the pre-response cost of the WTC (resp. LWS) must be smaller (resp. larger) than its pre-response cost, which can be seen by comparing the 3rd and 4th lines (resp. 9th and 10th

Table 6
Cost of the reforms, including behavioral responses

		France	Germany	Finland
<i>Working tax credit</i>				
Apparent cost	Billion euros/year	5.86	7.66	0.51
	% GDP	0.40%	0.36%	0.39%
Cost	Billion euros/year	5.74	6.22	0.49
Real cost (incl. labor supply response)	Billion euros/year	7.90	7.17	0.52
No. of recipient (hh)	% of population	10.1%	9.0%	10.6%
No. after response	Idem	11.2%	9.5%	10.7%
Net average amount per hh	(euros/month)	207	151	165
Net average amount after response	Idem	256	165	172
<i>Low-wage subsidy</i>				
Cost	Billion euros/year	8.30	7.52	0.52
Real cost (incl. labor supply response)	Billion euros/year	7.92	7.25	0.52
No. of recipient hh	% of population	22.8%	16.3%	21.9%
No. after response	Idem	23.2%	16.5%	22.0%
Average amount per hh	(euros/month)	132	101	84
Net average amount after response	Idem	124	96	83

Source: authors' computations using EUROMOD.

lines) of Table 6. As negative responses are larger in France, it appears that the real cost increases dramatically for this country once responses are accounted for (from 5.7 up to 7.9 billion EUR). Conversely, the real cost of the LWS is smaller than the pre-response cost (7.9 versus 8.3 billion EUR in France, for instance).

By the same token, the net average transfer of WTC increases after responses while the average amount of LWS decreases. With both reforms, the number of recipients increases, whether responses are positive (the rise in labor supply may open eligibility) or negative (the withdrawal of the woman from the labor market may cause the husband to become eligible). For instance, the number of households eligible to the WTC shifts from 9 to 9.5% in Germany and the number of households eligible to the LWS shifts from 22.8 to 23.2% in France.²³

In the case of the WTC, notice that the pre-response cost is lower than the *apparent cost* (first row in Table 6), i.e. the total amount of WTC distributed by the government. In effect, the introduction of the WTC partly crowds out spending on other transfers, in particular on social assistance.²⁴ In France, for instance, the difference between apparent (5.86 billion EUR) and net costs (5.74) is explained for 97% by the subsequent decrease in social assistance. The difference between apparent and net costs is especially large in Germany (around 20%), due to the generosity of the German minimum income scheme. In relative terms, the apparent cost of the WTC is slightly smaller in Germany (0.36% of 1998 GDP) than in France or Finland (around 0.40% of 1998 GDP). In absolute terms, the order of magnitude in France and Germany (5.86 and 7.66 billion Euros respectively) can be compared to the £5 billion (7.3 billion EUR) spent in 2001 in the UK on the WFTC, despite the fact that the UK reform does not concern childless households.

6.3. Targeting, distributional impact and cost efficiency

The targeting of the reforms, summarized in Table 6, reflects the intuitions from the analysis of framework conditions well. In effect, the number of recipients targeted by the WTC is larger in Finland (10.6% of the households) and France (10.1%) than in Germany (9%), mostly due to higher participation rates. Results are nevertheless reasonably comparable across countries. In contrast, the individual LWS reform is by nature less targeted; the number of recipients is in effect twice larger than in the case of the WTC. Important concentrations at the lower end of the income distribution in France and Finland explain a relatively larger number of eligible individuals in these countries. Differences in average amounts per household between countries are driven by a larger relative cost in France.

Fig. 5 shows that the WTC targets the first half of the distribution of equivalized incomes, just as in the UK. The first decile receives less than any of the three next deciles as it is composed mainly of inactive households which are not targeted by the reform. As seen in Fig. 5, the LWS attains individuals in all income deciles as low-wage individuals can be found in all families throughout the income distribution.

²³ Evidently, when labor supply responses are added to the picture, the number of recipients increases by less than the number of ‘movers’. Indeed, part of the movers were already entitled to the benefit before the transition. In Germany, this is the case for 23% (resp. 21%) of the movers due to the LWS (resp. WTC). These figures are 44% and 32% respectively in France and 15% and 12.5% in Finland.

²⁴ There is no difference between apparent and net costs in the case of the LWS as the reform does not interact with the rest of the system.

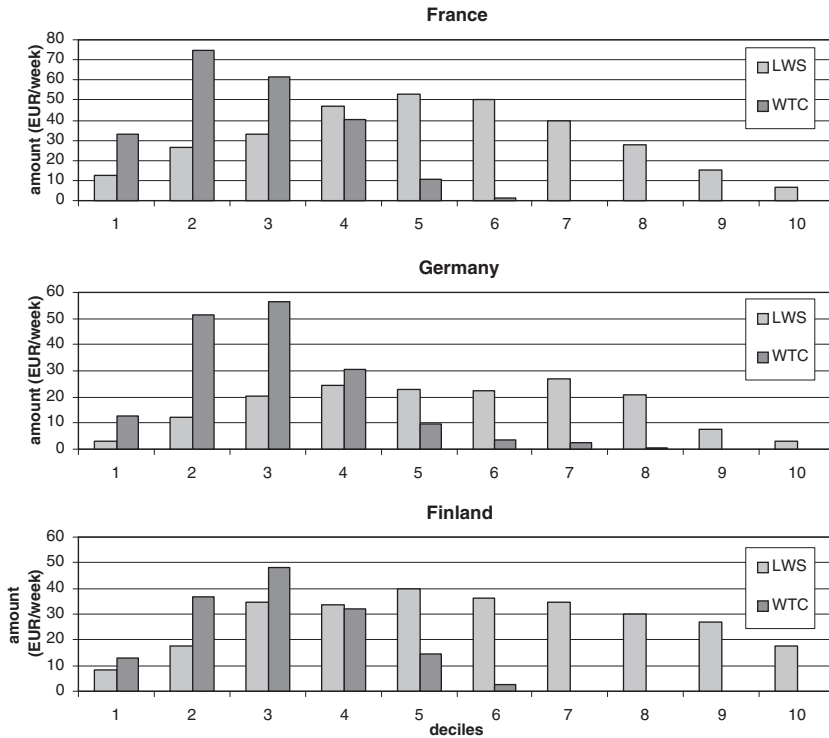


Fig. 5. Average amount of transfer per decile of adult equivalent disposable income.

Finally, we address a clear distributive policy objective, namely the *reduction of poverty*. We gauge the number of households taken out of poverty by holding the pre-reform poverty line constant, considering poverty lines defined as 40, 50 and 60% of the median of equalized disposable income.²⁵

Let us first ignore behavioral responses. Table 7 shows that both the WTC and the LWS achieve significant poverty reduction in France as the poverty rate declines from 7.03% to 6.38% (resp. 6.48%) with the WTC (resp. LWS), at the 50% poverty line. Surprisingly, the WTC succeeds only slightly better than the wage subsidy and even less when we consider the 40% line. In Germany, the reduction is not as large and the effects of both reforms are not significantly different. In Finland the reduction is even smaller than in Germany, unless the 60% poverty line is considered. Overall in all countries, poverty reduction is all the smaller as we consider lower poverty lines: the poorest among the poor are indeed most often inactive and do not benefit from the reforms.

A central question to our study is *whether increased labor participation is itself responsible for important moves across the poverty line*. Positive labor supply responses of single women (in the case of the WTC) and married women (with the LWS) indeed enhance poverty reduction to

²⁵ Mantovani and Sutherland (2001) compare poverty rates obtained using EUROMOD and national official figures. Figures are quite in line for France (the 1997 Fiscal Data give 6.9% with a poverty line at 50%) and Finland (the 1999 Income Distribution Survey gives 4%). Differences are somewhat larger in Germany (the 1999 GSOEP gives 8.2%), which is due to assuming 100% take-up of welfare assistance (Sozialhilfe). Notwithstanding, we are primarily concerned here with relative moves in and out of poverty than with the absolute level of headcount ratios.

Table 7
Distributive effects of the reforms

	Baseline (%)	WTC (%)	WTC + response (%)	LWS (%)	LWS + response (%)
<i>France</i>					
Poverty rate - line at 50% of the median	7.03	6.38	6.35	6.48	6.45
Variation in the number of poor hh (line at 60%)		–12.2	–12.4	–8.7	–9.3
Variation in the number of poor hh (line at 50%)		–9.3	–9.7	–7.9	–8.3
Variation in the number of poor hh (line at 40%)		–6.6	–6.6	–7.3	–7.3
<i>Germany</i>					
Poverty rate - line at 50% of the median	5.65	5.51	5.41	5.52	5.50
Variation in the number of poor hh (line at 60%)		–4.3	–5.6	–2.8	–3.3
Variation in the number of poor hh (line at 50%)		–2.5	–4.4	–2.3	–2.8
Variation in the number of poor hh (line at 40%)		–1.2	–3.7	–0.6	–1.2
<i>Finland</i>					
Poverty rate - line at 50% of the median	3.75	3.72	3.71	3.67	3.66
Variation in the number of poor hh (line at 60%)		–3.2	–3.6	–4.5	–4.9
Variation in the number of poor hh (line at 50%)		–0.7	–0.9	–2.1	–2.4
Variation in the number of poor hh (line at 40%)		–	–	–2.5	–2.5

Poverty line kept fixed at the baseline value.

some extent. Yet they do not dramatically change the picture in France or Finland. Things are markedly different in Germany. It turns out that in this country the number of households taken out of poverty by the WTC is almost doubled (resp. tripled) by behavioral responses when the 50% (resp. 40%) poverty line is considered. This is explained by the strong incentive effect of this reform on lone mothers who are more often in situation of poverty than in the two other countries (see the discussion in Section 6.1). Interestingly, then, generous in-work transfers as in a WFTC-type scheme may have a double dividend in reducing both social exclusion and poverty of this vulnerable group.

Finally, we study the cost efficiency of the reforms. We simply compute the cost per woman taking up work and the cost per household taken out of poverty. In Finland labor supply responses and distributive effects of both reforms are the smallest, which leads to extremely high efficiency costs. Costs are very high in the two other countries as well and markedly higher in Germany: it would cost 63,000 EUR in France and 90,000 EUR in Germany to bring a woman back to work through the LWS reform and respectively 53,000 (resp. 62,000) EUR and 131,000 (resp. 146,000) EUR to take a woman out of poverty by means of the WTC (resp. LWS). These figures seem large but remain within a scale similar to those observed in previous experiences. According to Pearson (2002) the cost per net job created has ranged between US \$30,000 and US \$100,000 throughout the various MWP policies experienced in the UK and the US.

7. Final discussion

In this paper, we present an extended analysis of ‘making work pay’ policies in Finland, France and Germany. These countries are characterized by very small incentives to work for low-wage households, the result of generous means-tested social transfers. Specifically, we simulate two in-work transfers in these countries. The first one is a working tax credit in the fashion of the British WFTC and the second is a simple wage subsidy. These two

reforms illustrate the typical opposition between family- and individual-based instruments in the current debate on ‘making work pay’ policies. Female labor supply estimations are carried out using homogeneous datasets and similar specifications across countries; the labor supply model is then combined with the integrated micro simulation EUROMOD in order to predict potential behavioral responses to the reforms. Differences in ‘framework conditions’ across countries are carefully analyzed, notably the differences in income and wage rate distributions, to explain the differences in the effects of each reform across countries.

We find that the overall female employment decreases after the introduction of the *working tax credit*. The participation of married women with children declines in all three countries and especially in France, where labor supply is slightly more elastic and where the net gain to work decreases substantially for secondary earners. This effect is only partially offset by a positive impact of the reform on lone mothers’ labor supply (and on a minority of childless women in couples in Germany). In contrast, in France and Germany, there are important efficiency gains to move toward the *individual wage subsidy* which, for the same fiscal cost, has an unambiguous positive effect on female employment. Results are more modest in Finland due to smaller labor supply elasticities of married women.

As far as *poverty reduction* is concerned, results are surprisingly not so different between the reforms. In fact, both reforms are work-conditional while the poorest households are to be found in the inactive population. This way, even if the WTC is better targeted at lower deciles (2,3,4) than the LWS, households of the first decile do not benefit much from either transfer. Overall, then, the individual policy measure seems more attractive insofar as it contributes both to social inclusion and poverty reduction. Note, however, that we have focused here on social inclusion as a whole, without distinguishing across household types. Yet, results show that the poorest group—single mothers—would be substantially induced to work thanks to the WTC in Germany. In this country, a generous transfer of the WTC-type could reach a double dividend by taking poor singles out of poverty and back to work at the same time. It seems reasonable to think that incentive objectives should focus more on the inclusion of single parents than that of a potential second-earner in couples. Ultimately, policy design could incorporate these aspects to generate beneficial effects for lone mothers while avoiding discouragement of secondary earners in couples.

Some improvements should be made to the present analysis. Firstly, a broader range of policy indicators could be used to differentiate the two instruments. Ultimately, the equity-efficiency trade-off that is central to the design of in-work programs should be addressed using a social welfare function. Further work is required in this direction and in particular a proper definition of the money metric equivalent to be used in coherence with the utility function. Also, the ‘social inclusion’ objective adopted here seems to have a social value per se in political debates—related to the fact that increased labor market participation generates positive social externalities as described by Phelps (2000)—which could not easily be incorporated into a social welfare function of the usual type.²⁶ Theoretical work is required to extend Mirrlees’ framework in several directions to eventually cover a broader range of policy concerns. We have focused here

²⁶ “Bringing marginalized groups, including those who work in the underground economy, into mainstream economic activities may generate beneficial outcomes for society as a whole, for example through the amelioration of problems like crime, social destitution, drug etc.”

on pragmatic policy criteria used by decision-makers and grounding the debates about reforms of European welfare systems.

Secondly, in the present work, the difference in tax-benefit environments has explained much of the differences in the effect of reforms across countries, in particular the difference in the generosity of the social assistance systems. It would be interesting to disentangle further explanatory factors when assessing the differences across countries, especially how much is due to different preferences versus different tax-benefit environments. A complementary exercise, left for future research, then consists of assuming identical elasticities across countries in order to capture what relates specifically to institutional factors.

Note finally that recent trends seem to somehow abandon the idea of expensive in-work transfers in favor of the ‘work first’ principle (claimants would be obliged to take job offers or risk a reduction of the benefit otherwise). However, it is not sure that the most stringent aspects of the recent *Hartz IV* reform in Germany be followed in practice and the new Minimum Income of Activity (*Revenu Minimum d'Activité*) is a non-compulsory alternative to social assistance in France.

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Appendix A

Table 8
Estimation results for women in couples

Variable	France		Germany		Finland	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
Income ²	−0.9182***	0.2283	−0.0708	0.1480	−0.6400*	0.3675
Female hours ²	26.7591***	4.8710	−10.4042	11.3407	88.9639***	21.2650
Female hours × income	0.0791	0.7971	−1.4011	0.9681	3.0467	2.2601
Income	6.6697	8.3859	4.8010	2.9404	7.4908*	4.5170
× female age/40	−14.9201	17.4067	−6.1830	4.1685	−10.0874	9.3105
× (female age/40) ²	13.8340	8.6703	2.5798	1.8804	5.7176	4.7403
× # children 0–2	1.7371	1.0652	−0.3860	0.3773	0.0990	0.5022
× # children 3–5	1.4826*	0.8256	−0.5646	0.4058	−0.1175	0.4818

(continued on next page)

Table 8 (continued)

Variable	France		Germany		Finland	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
× # children 6–11	1.4258**	0.5577	−0.2772*	0.1673	0.2832	0.3458
× # children 12+	−0.1464	0.4774	−0.2937	0.1795	0.4706	0.3532
× 1(region) [@]	1.2552	0.8146	1.1588**	0.5487	0.4981	0.5482
Female hours	−24.9799**	10.0216	21.0038*	12.2902	−73.8700***	22.7041
× female age/40	54.7356***	20.7632	−6.7550	17.3435	65.2742*	36.6370
× (female age/40) ²	−38.9307***	10.7158	−2.4596	8.1034	−33.9415*	17.7977
× # children 0–2	−2.8299*	1.6839	−4.6706**	2.2478	−6.3646**	3.0145
× # children 3–5	−3.9432***	1.3632	−9.5222***	1.6166	−0.6668	2.8390
× # children 6–11	−3.9210***	0.7678	−6.1076***	0.7095	−1.5971	1.3800
× # children 12+	−1.2452**	0.5531	−2.5909***	0.5576	−1.8834	1.2544
× 1(region) [@]	0.7933	1.0009	8.9317***	1.3154	−0.9320	1.8412
Fixed cost	0.0229	0.0742	1.3190*	0.7002	−1.2076	0.8220
× # children 0–2	0.1943**	0.0789	2.2509*	1.2467	−0.1601	0.3140
× # children 3–5	0.0166	0.0562	0.8150	0.6306	−0.4426	0.4137
× 1(lower education)	0.1420***	0.0473	1.4069	1.2516	0.1006	0.1001
Log–Likelihood	−1667		−917		−996	
No. of observations	2095		1265		1632	
pseudo-R2	27.6%		34.0%		44.5%	

Level of significance: * = 10%, ** = 5%, *** = 1%.

[@]The dummy 'region' corresponds to Paris area for France, Helsinki area for Finland and East Germany for Germany.

Table 9

Estimation results for single women

Variable	France		Germany		Finland	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
Income ²	−3.0837**	1.2504	−8.2232***	2.2528	−3.6534	2.9963
Female hours ²	1.3856	9.1544	14.4934*	7.9867	89.0400***	12.5828
Female hours × income	12.0887**	5.8195	1.7016	7.0759	3.8994	6.9608
Income	−8.7200	12.5077	−0.3680	11.1618	−3.9123	15.1474
× female age/40	13.9328	23.7001	15.0328	22.9343	10.3076	30.4375
× (female age/40) ²	−4.9919	10.9727	−7.2789	10.7651	−2.4654	14.2924
× # children	0.2144	0.8686	−0.3091	0.8341	0.9738	1.4088
× 1(region) [@]	1.4111	2.0674	−1.1437	1.6278	−1.5189	1.7070
Female hours	−1.7539	14.1947	−164.9638***	9.5845	−63.3330***	15.4236
× female age/40	13.9328	27.9240	36.5580**	17.7588	60.5406**	30.2555
× (female age/40) ²	−10.9208	12.9809	−20.6310**	8.3226	−29.1719**	14.4698
× # children 0–2	−5.0343***	1.8877	−18.0533***	3.8716	−6.7545**	3.2941
× # children 3–5	−3.5248**	1.6948	−12.9142***	2.1358	−2.3168	1.7466
× # children 6–11	−4.7025***	1.1960	−5.5291***	0.9490	−1.8252	1.4415
× # children 12+	−1.9553	1.2528	−3.2961	0.9179	−1.3726	1.5842
× 1(region) [@]	0.6284	2.4008	4.1984***	1.4068	2.6823	1.6776
Fixed cost	0.1253	0.2265	0.4086**	0.1604	ns	ns
Log–Likelihood	−299		−247		−209	
No. of observations	664		453		416	
Pseudo-R2	59.1%		50.4%		54.2%	

Level of significance: * = 10%, ** = 5%, *** = 1%.

ns: covariates are excluded from the estimation if they are highly non-significant, hence increasing the variance of the predictions from the estimated model.

[@]The dummy 'region' corresponds to Paris area for France, Helsinki area for Finland and East Germany for Germany.

Table 10
Goodness-of-fit for women in couples

		France			Germany			Finland		
Choice		Observed frequencies	Average predicted frequencies	Generalized R2 (%)	Observed frequencies	Average predicted frequencies	Generalized R2 (%)	Observed frequencies	Average predicted frequenciesz	Generalized R2 (%)
Married women	0	0.296	0.300	18.6	0.368	0.368	39.0	0.264	0.284	25.5
	Part-time	0.117	0.114	0.6	0.166	0.166	2.7	0.038	0.041	0.1
	Full-time	0.588	0.586	18.2	0.466	0.466	34.6	0.698	0.675	23.5
Single women	0	0.032	0.032	6.7	0.208	0.208	43.2	0.200	0.215	16.8
	Part-time	0.117	0.113	2.7	0.093	0.093	4.1	0.019	0.012	0.1
	Full-time	0.851	0.855	7.0	0.700	0.700	37.5	0.781	0.773	16.2

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